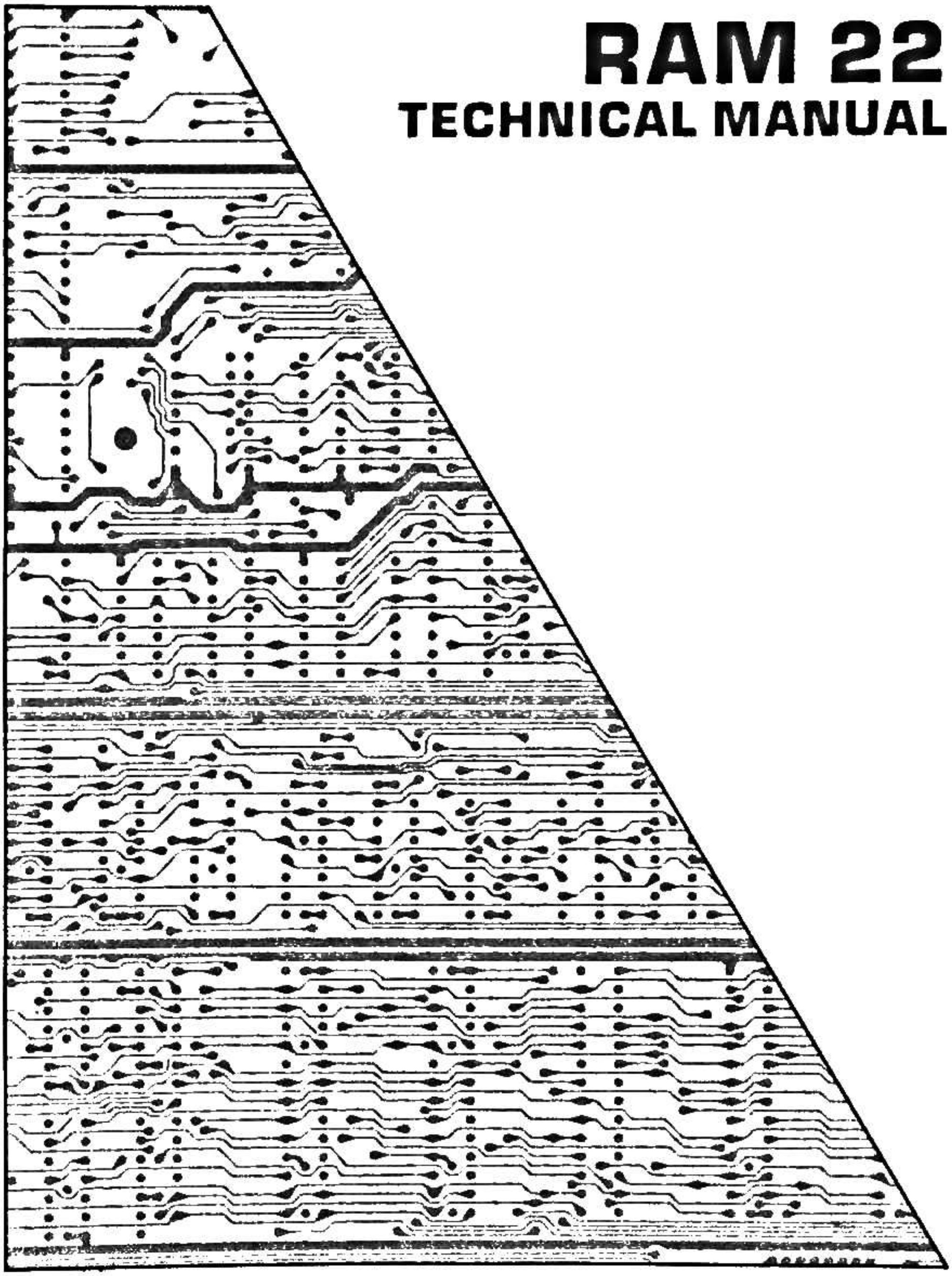
CompuPro™

RAM 22
TECHNICAL MANUAL



8261-0051B

May 1987

RAM 22
TECHNICAL MANUAL

256 KByte Static Memory
for the IEEE 696/S-100 Bus

RAM 22 Technical Manual ASY-198C
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Specifications

Timing	Meets all IEEE 696/S-100 specifications
Speed	RAM chip access time 120 ns.
Address Bits	24-bits; meets IEEE 696/S-100 extended address specifications
Address	DIP switch selectable on any 256K byte boundary.
Data Transfer	8-bit or 16-bit; conforms with IEEE 696/S-100 timing requirements for sXTRQ* and SIXTN*
Power Consumption	Less than 4 watts typical at nominal +8V

About the RAM 22

The RAM 22 from **CompuPro** represents one of the most advanced RAM boards ever produced for the IEEE 696/S-100 Bus. By combining state-of-the-art static CMOS RAM technology with **CompuPro's** design excellence, the RAM 22 offers the most versatile, efficient and reliable performance available today. The board works as "word-wide" memory for today's 16-bit systems and automatically switches to "byte-wide" mode in 8-bit systems.

The RAM 22 uses 32 high performance 8K X 8 CMOS RAM chips to provide a total of 256 Kbytes or 128 Kwords of storage. The RAM 22 is addressable on any 256 Kbyte boundary in the 16 megabyte address space specified by the IEEE 696 standard.

CompuPro's RAM 22 will work with all CompuPro CPUs at their highest speeds (up to 12.5 MHz). Refer to the software Installation Guide for the appropriate settings on the CPU, and/or the notes on **Using the RAM 22 with the High Speed 286 CPUs** in this manual.

The RAM 22 also handles high-speed DMA flawlessly, a feature few memory boards can boast.

CompuPro also uses a high-quality double-sided circuit board design that has a full solder-mask and legend. Sockets are provided for all ICs for ease of maintenance. All edge connector contacts are gold on a nickel substrate to ensure long and reliable operation.

NOTE: This board does not allow any of the four 64K blocks on the board to be disabled.

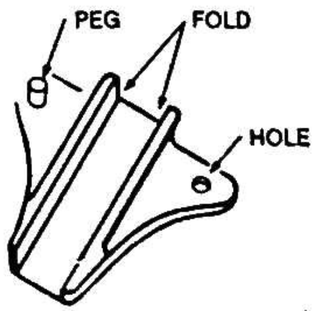
Installing the RAM 22 Board

Basic Installation

Step 1. Unpack the RAM 22 Board.

Along with the board, you will find two card extractors in the plastic bag.

CARD EXTRACTOR



Step 2. Install Card Extractors.

1. Hold the board so the component side is toward you.

2. Insert the peg on the card extractor into the hole in the right corner of the board. Fold the extractor over the board's edge until the extractor's hole snaps over the peg (make sure the long edge of the card extractor is along the top edge of the board.)

3. Repeat for the left extractor.

Step 3. Check Switch and Jumper Settings.

For standard switch settings for a CompuPro operating system check the operating system Installation Guide. Otherwise, refer to **Switch Settings** and **Jumper Settings** in this manual.

Step 4. Insert the RAM 22 into the S-100 Bus.

The power to the system must be off. Place the board into a slot towards the front of the enclosure. The edge connector is offset, so the RAM 22 will fit only one way. **Push down GENTLY until the board is firmly installed.**

Switch Settings

S1-1 (Switch 1 paddle 1)

There is no connection to this paddle. Set it OFF.

S1-2 through S1-7 - Address Select

The starting address of the board is selected by setting the middle six paddles of dip-switch S1. The address is set in a binary fashion with each paddle of S1 representing an address bit. An "ON" paddle represents a binary "zero" and an "OFF" paddle represents a binary "one". Paddle 2 is the MSB and corresponds with A23 and paddle 7 is the LSB and corresponds to A18.

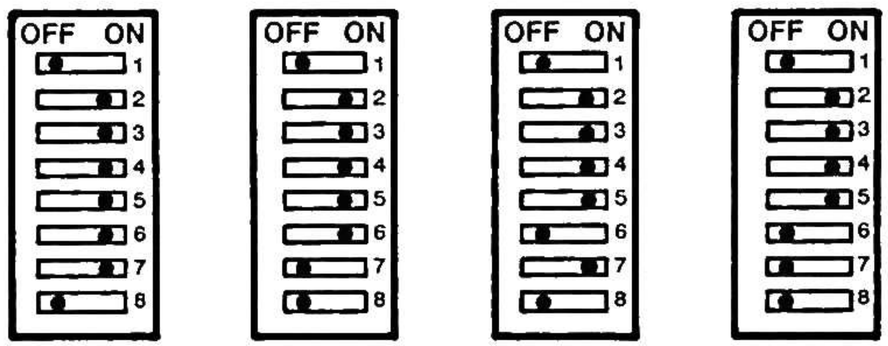
S1-8 (Switch 1 paddle 8)

There is no connection to this paddle. Set it OFF.

Sample Switch Settings

The following are the four possible switch settings for the 1 Mbyte range addressable by the CPU 8086 and CPU 8085/88:

000000h	040000h	080000h	0C0000A



Jumper Settings

J1 - Memory Scheme Select

This is almost always installed. When the shunt is removed, the lower 16 Kbytes of every 64 Kbyte address range becomes global memory.

Using the RAM 22 with High Speed 286 CPUs

Revision 451 or newer of the CPU 286[tm] may be too fast for the RAM 22 if J1 on the CPU 286 board is jumpered for no memory waits (two-cycle operation).

The RAM 22 Revision C can support the following:

- A 10 Mhz or 12.5 Mhz CPU 286 Revision 451 with one wait state in all cycles
- An 8 MHz CPU 286 Revision 451 with no memory wait states; one wait state in all other cycles
- An 8 Mhz CPU 286 Revision 201 with one wait state in all cycles
- A 6 MHz CPU 286 Revision 201 with no memory wait states; one wait state in all other cycles

Theory of Operation

The RAM 22 is designed to work in 8- and 16-bit systems per the protocol established by the IEEE 696/S-100 standard. The DATA IN and DATA OUT buses operate as a bidirectional 16-bit data path when word transfers are performed. The two buses remain uni-directional during byte operations.

The RAM 22 handles this multiplexing of the data buses with the two bi-directional bus buffers U5 and U11 (both 74LS245) and the intermediate buffer U8 (also 74LS245). Both U5 and U11 are enabled for all word transfers. Both U5 and U11 are enabled for word read operations. When byte writes occur with A0=0, U15 is enabled. When byte writes occur with A0=1, U5 and U8 are enabled. When byte reads occur with A0=0, U8 and U11 are enabled. When byte reads occur with A0=1, U11 is enabled.

This algorithm is executed by PAL (programmable array logic) 329 (U9). A second PAL 328 (U4) acts as the array decoder along with U3. The RAM is configured as two arrays of 128K by 8 bits, A and B. The bit/address decode scheme is covered in the next section. The decoder PAL only generates select signals, which enable the RAM chips only during memory reference operations. This feature, coupled with the use of RAM chips which power down when not selected, makes the RAM 22 consume less power than dynamic RAM designs while providing the speed of operation and reliability that only static RAM delivers.

The base address of the board is set with dip-switch S1. Octal comparator (U8) generates signal ASEL* when the address present on bus lines A18 through A23 matches that set in S1-2 through S1-7. Positions 1 and 8 of S1 are not used.

ESXT* is the signal generated in the PAL which causes the RAM 22 to acknowledge requests for word transfers when the board is selected. Transistor Q1 provides the open collector output required to drive bus signal SXTN* (line 60).

Locating RAM ICs by Address and Bit

The **Component Layout** at the back of this manual may be used as a map to locate RAM ICs by address and byte.

Each 6264 RAM chip is 8 Kilobytes. By decoding A0, one is selected for an even byte, another for an odd byte. This gives 16 Kbytes. By decoding A15 and A14 a 64 Kbyte page can be decoded into 16K divisions. Finally decoding A17 and A16 allows 256 Kbytes of memory to be divided into 64 Kbyte pages.

The eight RAM chips at the center of the board (0B0-0A3 in the component layout diagram at the rear of the manual) make up the zero page of memory. The even bytes are the lower row and the odd bytes are above. The first page of memory starts a regular pattern that is repeated for the following two pages. It is comprised of the 4 chips at the lower left (even bytes) and the 4 chips at the lower right (odd bytes) of the board. The 8 chips half-way up the left and right sides of the board are page 2. Finally the page 3 chips are at the top left and right of the board.

Within each group of 4 chips making up a given even or odd page, their address increases by 16K going right to left.

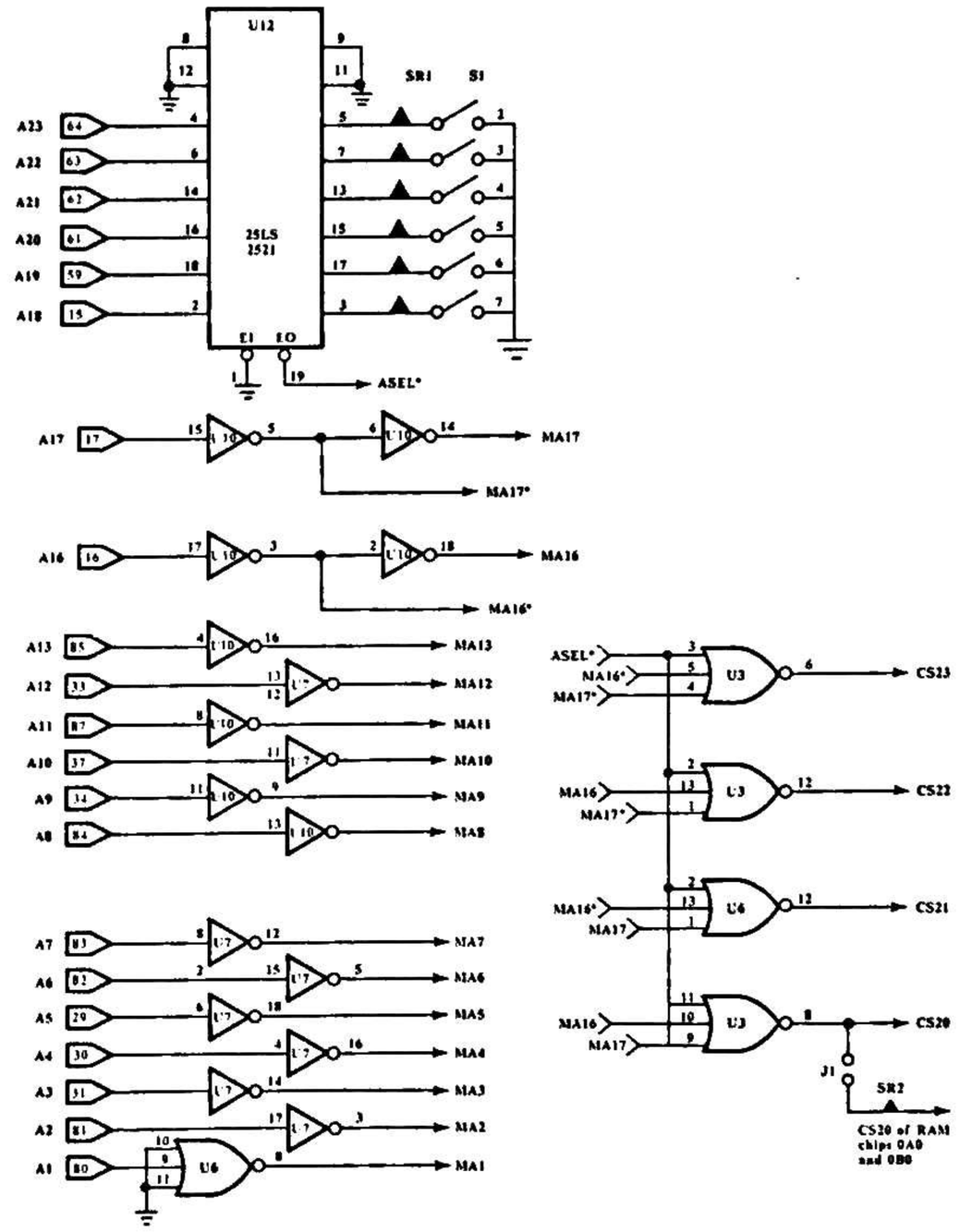
There is a second method of locating RAM chips. Each chip is labeled xyz e.g. 0A0, 2A3, etc. Let x = the decimal value of the binary pair A17 and A16. Let y = A for A0 = 0 (even bytes) and B for A0 = 1 (odd bytes). Let z = the decimal value of the binary pair A15 and A14. Now with a given address we can find which chip (or pair of chips if a word is being accessed) is being selected.

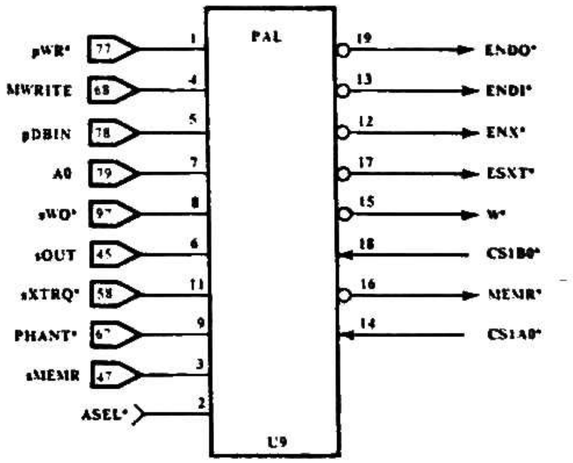
For example: Suppose you wish to find the chip on a RAM 22 addressed at 080000h through 0BFFFFh that responds to address 0A5216h.

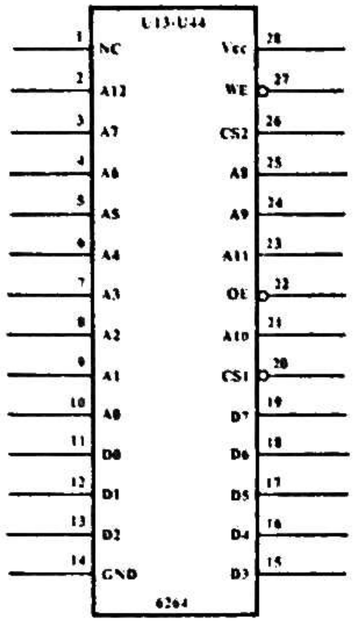
With	Then
A17 = 1 and A16 = 0	x = 2
A15 = 0 and A14 = 1	z = 1
A0 = 0	y = A

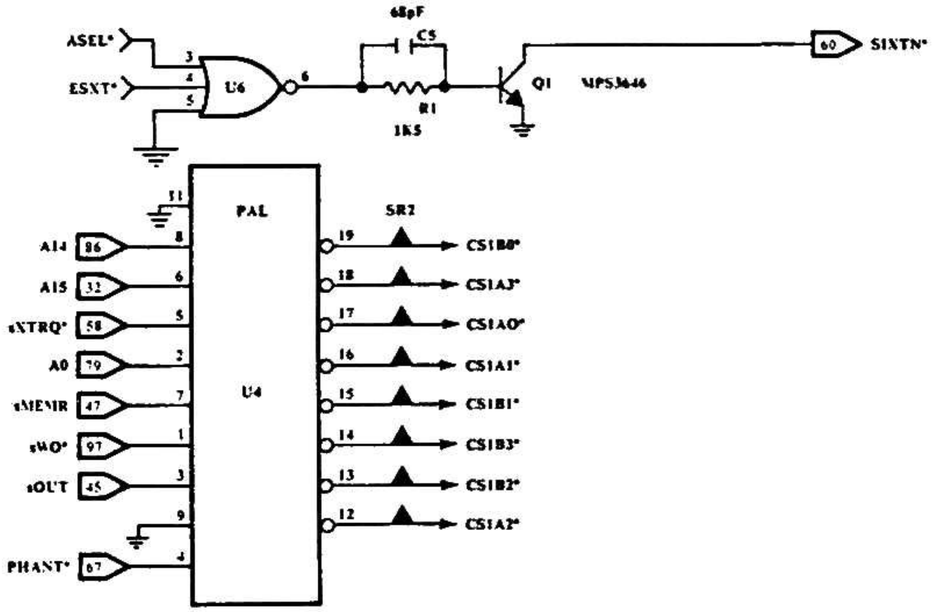
Thus

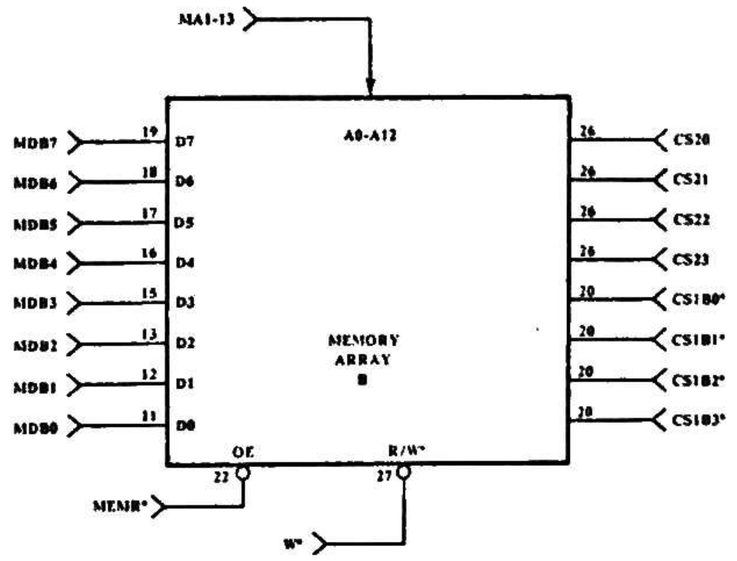
xyz = 2A1 (middle row, third from right)

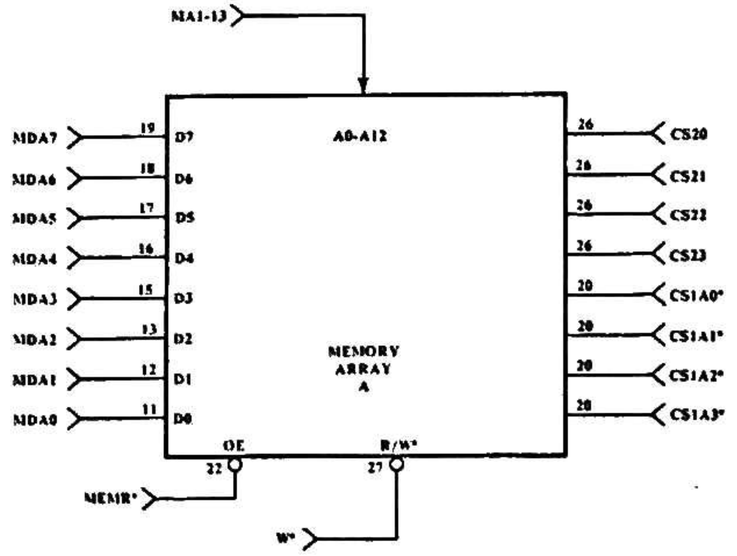


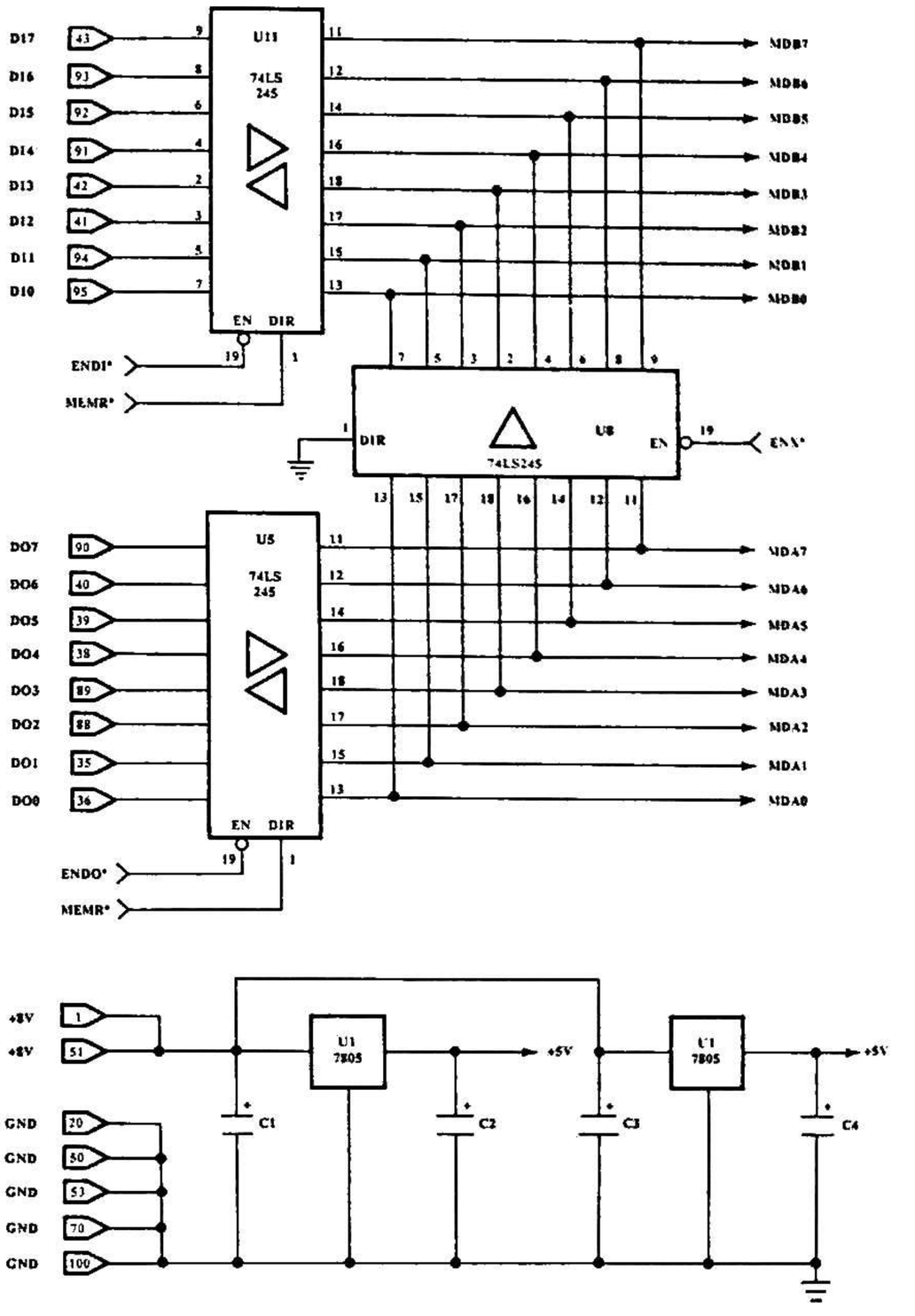

PAL
pWR*
MWRITE
pDBIN
A0
sWO*
sOUT
sXTRQ*
PHANT*
sMEMR
ASEL*
ENDO*
ENDI*
ENX*
ESXT*
W*
CS1B0*
MEMR*
CS1A0*
U9

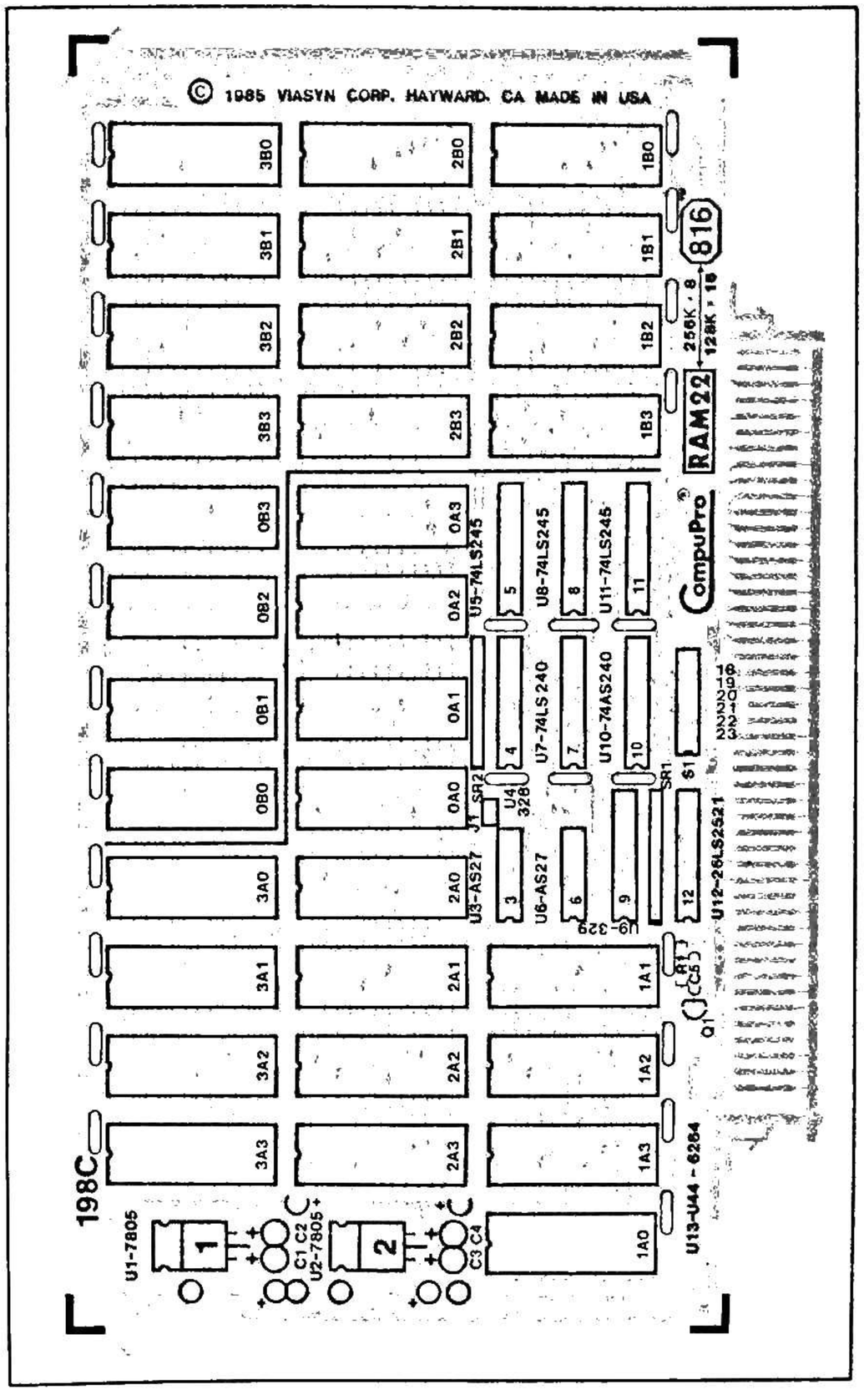

U13-U44
NC
A12
A7
A6
A5
A4
A3
A2
A1
A0
D0
D1
D2
GND
Vcc
WE
CS2
A8
A9
A11
OE
A10
CS1
D7
D6
D5
D4
D3
6264


ASEL*
ESXT*
U6
68pF
C5
R1
1K5
Q1
MPS3646
SIXTN*
PAL
SR2
A14
A15
sXTRQ*
A0
sMEMR
sWO*
sOUT
PHANT*
U4
CS1B0*
CS1A3*
CS1A0*
CS1A1*
CS1B1*
CS1B3*
CS1B2*
CS1A2*


MA1-13
A0-A12
MDB7
MDB6
MDB5
MDB4
MDB3
MDB2
MDB1
MDB0
D7
D6
D5
D4
D3
D2
D1
D0
MEMORY
ARRAY
B
OE
R/W*
MEMR*
W*
CS20
CS21
CS22
CS23
CS1B0*
CS1B1*
CS1B2*
CS1B3*


MA1-13
A0-A12
MDA7
MDA6
MDA5
MDA4
MDA3
MDA2
MDA1
MDA0
D7
D6
D5
D4
D3
D2
D1
D0
MEMORY
ARRAY
A
OE
R/W*
MEMR*
W*
CS20
CS21
CS22
CS23
CS1A0*
CS1A1*
CS1A2*
CS1A3*





Component Layout

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Viasyn Corporation warrants this computer product to be in good working order for a period of 180 days from the date of shipment from the factory, or 90 days from the date of retail sale of the product to the original end user, whichever comes first. Should this product fail to be in good working order at any time during this warranty period, VIASYN will, at its option, repair or replace the item at no additional charge except as set forth below. Repair parts and replacement products will be furnished on an exchange basis and will be either reconditioned or new. All replaced parts and products become the property of VIASYN. This limited warranty does not include service to repair damage to the product resulting from accident, disaster, misuse, abuse or unauthorized modification of the product. To obtain service under this warranty the item must be returned to VIASYN as described below.

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